Appointment Management — Use Case Documentation

Project: Biomat USA Donor Management and Appointment Scheduling
Course: IS 6465 — Web Based Applications
Team: Caleb Syndergaard, Josh Stott, Cade Bray
Section author: Josh Stott (Appointment Management)

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1. Overview

This document describes the four Appointment Management use cases that make up the appointment CRUD set in the team proposal. Each description reflects the behavior of the existing application (the My appointments page, appointments.php). Authentication is treated as a precondition, so Log In is not described as a separate use case here.

ID	Use Case	CRUD	Summary
UC-A1	Request Appointment	Create	Donor books a new appointment at an active donation center.
UC-A2	View Appointment History	Read	Donor views all of their own past and upcoming appointments.
UC-A3	Update Appointment	Update	Donor changes the center or date and time of an upcoming booked appointment.
UC-A4	Cancel Appointment	Delete (soft)	Donor cancels an upcoming booked appointment; the record is kept with status Cancelled.

2. Terminology

These terms are used the same way in all four use cases.

Term	Meaning
Donor	A registered user signed in with a donor account that is linked to a donor profile.
System	The Biomat Donor Services web application.
Appointments page	The My appointments page, opened from Appointments in the navigation bar. It holds the appointment form and the Appointment history.
Appointment form	The form on the Appointments page. It is titled Request an appointment when creating, and Update appointment when editing.
Appointment history	The list of the Donor's appointments on the Appointments page.

Term	Meaning
Active donation center	A donation center marked active. Only active centers appear in the Donation center list.
Appointment status	Booked (scheduled), Cancelled (cancelled by the Donor), or Completed (set by personnel in the Staff workspace).
Upcoming booked appointment	An appointment that belongs to the Donor, has status Booked, and has a start date and time later than the current time. Only these can be updated or cancelled.
Date and time (UTC)	All appointment dates and times are entered and displayed in Coordinated Universal Time.

UC-A1: Request Appointment (Create)

UC-A1: Request Appointment	
Use Case ID	UC-A1
Use Case Name	Request Appointment
Primary Actor	Donor
CRUD Operation	Create
Description	The Donor requests a new plasma donation appointment by selecting an active donation center and a future date and time. The System validates the request and saves the appointment with status Booked.
Preconditions	**1.**The Donor is logged in with a donor account linked to a donor profile. **2.**At least one active donation center exists in the System.
Trigger	The Donor selects Appointments in the navigation bar to request a new appointment.
Main Success Scenario	**1.**The Donor selects Appointments in the navigation bar. **2.**The System displays the Appointments page with the Request an appointment form and the Appointment history. **3.**The Donor selects a donation center from the Donation center list, which shows only active donation centers. **4.**The Donor enters an appointment date and time (UTC). **5.**The Donor clicks Request appointment. **6.**The System validates the donation center: a center is selected and the center is active. **7.**The System validates the date and time: the value is a valid date and time and is later than the current time. **8.**The System saves the appointment for the Donor and the selected center with status Booked. **9.**The System redisplays the Appointments page with the message "Your appointment request has been saved." The new appointment appears in the Appointment history with Edit and Cancel actions.

UC-A1: Request Appointment	
Alternative / Exception Flows	**Exception flows** **1a.Donor is not logged in.** The System redirects the Donor to the Sign in page and displays "Please sign in to continue." The use case ends. **5a.A required field is empty.** The browser blocks submission and prompts the Donor to complete the field. The flow returns to step 3. **5b.Form security token is invalid or expired.** The System rejects the request and displays "The form expired. Please go back, reload the page, and try again." No data is changed. The use case ends. **Validation flows** **6a.No center is selected.** The System displays "Choose a donation center." **6b.The selected center is not active.** The System displays "Choose an available donation center." **7a.The date and time is invalid or not in the future.** The System displays "Choose a valid appointment date and time in the future." For 6a–7a, the System completes all checks, lists every error message together at the top of the form, keeps the Donor's entered values, and saves nothing. The flow returns to step 3.
Postconditions	**Success** A new appointment record exists for the Donor with the selected center, date and time, and status Booked. It appears in the Appointment history. **Failure** No appointment record is created.
Business Rules	• A Donor can create appointments only for themselves. • Only active donation centers can be selected. • The appointment date and time must be later than the current time. • The System does not check time-slot availability or prevent overlapping appointments.
Related Use Cases	The new appointment appears in UC-A2 View Appointment History and can then be changed (UC-A3) or cancelled (UC-A4).

UC-A2: View Appointment History (Read)

UC-A2: View Appointment History	
Use Case ID	UC-A2
Use Case Name	View Appointment History
Primary Actor	Donor
CRUD Operation	Read
Description	The Donor views a list of all of their own appointments, past and upcoming, including each appointment's status, center and date and time. Upcoming booked appointments show actions to update or cancel them.
Preconditions	**1.**The Donor is logged in with a donor account linked to a donor profile.
Trigger	The Donor selects Appointments in the navigation bar. The System also shows the Appointment history after each appointment request, update or cancellation.
Main Success Scenario	**1.**The Donor selects Appointments in the navigation bar. **2.**The System retrieves all appointments that belong to the logged-in Donor. **3.**The System sorts the appointments by date and time, most recent first. **4.**The System displays each appointment in the Appointment history with a date badge (month and day), status (Booked, Cancelled or Completed), donation center name, date and time (UTC), and center city and state. **5.**For each upcoming booked appointment, the System displays Edit and Cancel actions. **6.**The Donor reviews the Appointment history.
Alternative / Exception Flows	**Exception flows** **1a.Donor is not logged in.** The System redirects the Donor to the Sign in page and displays "Please sign in to continue." The use case ends. **2a.The Donor has no appointments.** The System displays "No appointments yet" with the note "Your upcoming and past appointments will appear here." The use case ends. **Alternative flows** **5a.The appointment is not an upcoming booked appointment** (its start time has passed, or its status is Cancelled or Completed). The System displays it without Edit and Cancel actions. The flow continues at step 6. **6a.The Donor clicks Edit on an appointment.** Continue with UC-A3 Update Appointment. **6b.The Donor clicks Cancel on an appointment.** Continue with UC-A4 Cancel Appointment.
Postconditions	**Success** The Donor's appointments are displayed. No data is changed.

UC-A2: View Appointment History	
Business Rules	• A Donor sees only their own appointments. • Cancelled appointments remain visible in the history. • Only upcoming booked appointments can be updated or cancelled. • The Completed status is set by personnel in the Staff workspace, not by the Donor.
Related Use Cases	Extended by UC-A3 Update Appointment and UC-A4 Cancel Appointment.

UC-A3: Update Appointment (Update)

UC-A3: Update Appointment	
Use Case ID	UC-A3
Use Case Name	Update Appointment
Primary Actor	Donor
CRUD Operation	Update
Description	The Donor changes the donation center or the date and time of an upcoming booked appointment. The System validates the changes and updates the existing appointment record.
Preconditions	**1.**The Donor is logged in with a donor account linked to a donor profile. **2.**The Donor has at least one upcoming booked appointment. **3.**The Donor is viewing the Appointment history (UC-A2).
Trigger	The Donor clicks Edit on an upcoming booked appointment in the Appointment history.
Main Success Scenario	**1.**The Donor clicks Edit on an upcoming booked appointment. **2.**The System verifies that the appointment is an upcoming booked appointment that belongs to the Donor. **3.**The System displays the Update appointment form, pre-filled with the appointment's current donation center and date and time. **4.**The Donor changes the donation center, the date and time, or both. **5.**The Donor clicks Save changes. **6.**The System validates the donation center: a center is selected and the center is active. **7.**The System validates the date and time: the value is a valid date and time and is later than the current time. **8.**The System confirms the appointment is still an upcoming booked appointment that belongs to the Donor, then saves the new center and date and time. The status remains Booked. **9.**The System redisplays the Appointments page with the message "Your appointment has been updated." The Request an appointment form is shown again, and the Appointment history shows the updated details.

UC-A3: Update Appointment	
Alternative / Exception Flows	**Exception flows** **1a.Donor is not logged in.** The System redirects the Donor to the Sign in page and displays "Please sign in to continue." The use case ends. **2a.The appointment cannot be changed** (its start time has passed, its status is not Booked, or it belongs to another donor). The System redisplays the Appointments page with the message "Only your upcoming booked appointments can be changed." The use case ends. **5b.Form security token is invalid or expired.** The System rejects the request and displays "The form expired. Please go back, reload the page, and try again." No data is changed. The use case ends. **8a.The appointment is no longer an upcoming booked appointment at save time** (for example, its start time passed while the Donor was editing). The System displays "This appointment is no longer available to update." and saves nothing. The use case ends. **Alternative flows** **4a.The Donor clicks Cancel edit.** The System discards the changes and displays the Appointments page with the Request an appointment form. The appointment is unchanged. The use case ends. **5a.A required field is empty.** The browser blocks submission and prompts the Donor to complete the field. The flow returns to step 4. **Validation flows** **6a.No center is selected.** The System displays "Choose a donation center." **6b.The selected center is not active.** The System displays "Choose an available donation center." **7a.The date and time is invalid or not in the future.** The System displays "Choose a valid appointment date and time in the future." For 6a–7a, the System lists every error message together, keeps the Update appointment form open with the Donor's entered values, and saves nothing. The flow returns to step 4.
Postconditions	**Success** The existing appointment record (same appointment ID) has the new center and/or date and time, and its status is still Booked. **Failure** The appointment is unchanged.
Business Rules	• Only upcoming booked appointments that belong to the Donor can be updated. • The Donor can change only the donation center and the date and time; the Donor cannot change the status. • The same validation rules as UC-A1 apply. • Updating changes the existing record; it does not create a new appointment.
Related Use Cases	Extends UC-A2 View Appointment History. Uses the same validation rules as UC-A1 Request Appointment.

UC-A4: Cancel Appointment (Soft Delete)

UC-A4: Cancel Appointment	
Use Case ID	UC-A4
Use Case Name	Cancel Appointment
Primary Actor	Donor
CRUD Operation	Delete (soft delete)
Description	The Donor cancels an upcoming booked appointment. This is a soft delete: the System changes the appointment's status to Cancelled and keeps the record in the Appointment history. The appointment is not permanently deleted.
Preconditions	**1.**The Donor is logged in with a donor account linked to a donor profile. **2.**The Donor has at least one upcoming booked appointment. **3.**The Donor is viewing the Appointment history (UC-A2).
Trigger	The Donor clicks Cancel on an upcoming booked appointment in the Appointment history.
Main Success Scenario	**1.**The Donor clicks Cancel on an upcoming booked appointment. **2.**The System displays a confirmation dialog: "Cancel this appointment?" with OK and Cancel buttons. **3.**The Donor clicks OK. **4.**The System verifies that the appointment is an upcoming booked appointment that belongs to the Donor. **5.**The System changes the appointment's status from Booked to Cancelled. The record is kept. **6.**The System redisplays the Appointments page with the message "Your appointment has been cancelled." The appointment appears in the Appointment history with status Cancelled and without Edit and Cancel actions.
Alternative / Exception Flows	**Exception flows** **1a.Donor is not logged in.** The System redirects the Donor to the Sign in page and displays "Please sign in to continue." The use case ends. **3b.Form security token is invalid or expired.** The System rejects the request and displays "The form expired. Please go back, reload the page, and try again." No data is changed. The use case ends. **4a.The appointment is no longer an upcoming booked appointment** (for example, its start time has passed or it was already cancelled). The System makes no change and displays "That appointment could not be cancelled. Refresh and try again." The use case ends. **Alternative flows** **3a.The Donor clicks Cancel in the confirmation dialog.** The dialog closes, nothing is sent to the System, and the appointment is unchanged. The use case ends.

UC-A4: Cancel Appointment	
Postconditions	**Success** The appointment record still exists with status Cancelled. It remains in the Appointment history and can no longer be updated or cancelled. **Failure** The appointment is unchanged.
Business Rules	• Only upcoming booked appointments that belong to the Donor can be cancelled. • The Donor must confirm the cancellation before the System changes the status. • Cancellation is a status change, not a permanent deletion (see Section 3). • A cancelled appointment cannot be reactivated; the Donor requests a new appointment instead (UC-A1).
Related Use Cases	Extends UC-A2 View Appointment History. A new appointment can be requested with UC-A1.

3. Cancellation vs. Permanent Deletion

Cancelling an appointment (UC-A4) is a soft delete. The System changes the appointment's status from Booked to Cancelled and keeps the record, so it stays in the Donor's Appointment history and remains available for reporting.

Donors cannot permanently delete an individual appointment. An appointment record is permanently removed only in two cases, both outside the scope of these use cases:

- The Donor deletes their own account (Donor Management), which removes the account's associated appointment and donation records.
- Personnel or an administrator deletes the appointment record in the Staff workspace.